👩‍🏫 **Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.**

UNIT 6 · STANDARD 6.NOS.1

Divide Mixed Numbers

Lesson 6-10 · Teacher Copy

Name: _____ **Date:** _____ **Class:** _____

My Notes

Level 3 Enrichment

🎯 **TODAY'S OBJECTIVES**

Content Objective: I can explain what dividing with mixed numbers means, rewrite them as improper fractions, find the quotient with common denominators or by multiplying by the reciprocal, and check it by multiplying.

Language Objective: I can explain my steps and why they work, using the words mixed number, improper fraction, convert, reciprocal, and simplify.

📖 See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.

✏️ **Fill in each blank as we go. Use the Word Bank to help you.**

📚 **WORD BANK — FILL EACH BLANK WITH THE BEST WORD**

Mixed Number | **Improper Fraction** | **Convert** | **Simplify** | **Reciprocal**

Level 3: try the blanks from memory first, then check the bank.

1 A number with a whole number and a fraction, like 2 1/3, is a _____.

2 A fraction whose numerator is greater than or equal to its denominator, like 7/3, is an _____.

3. Changing a mixed number into an improper fraction is to _____ it.

4. Writing a fraction with smaller numbers that names the same amount, like 4/8 = 1/2, is to _____ it.

5. To divide by a fraction, I multiply by its _____ .

Write About the Math

How does dividing mixed numbers help the detective?
¿Cómo ayuda dividir números mixtos a la detective?

Say your answer to a partner first. Then write it, using at least two of these words:

- ☐ **Mixed Number** — *Número mixto*
- ☐ **Improper Fraction** — *Fracción impropia*
- ☐ **Convert** — *Convertir*
- ☐ **Simplify** — *Simplificar*
- ☐ **Reciprocal** — *Recíproco*

Model: *A mixed number must first become an improper fraction so I can flip and multiply.* — Students explain that the mixed number must be converted to an improper fraction (7/2) before applying the reciprocal shortcut.

Start

Complete the frames. Use the word bank.

Completa las oraciones. Usa el banco de palabras.

- **The detective can cut ___ pieces because 5 1/4 ÷ 3/4 = 21/4 × 4/3 = 84/12 = ___.**
- **My answer is ___ because ___.**
  Mi respuesta es ___ porque ___.

Build

Write two connected sentences. Use because, so, or but.

Escribe dos oraciones conectadas. Usa porque, entonces o pero.

- **My claim is ___.**
  Mi afirmación es ___.
- **My evidence is ___, so ___.**
  Mi evidencia es ___, entonces ___.

Explain

Write a claim, give math evidence, and explain your reasoning.

Escribe una afirmación, da evidencia matemática y explica tu razonamiento.

- [ ] I answered the question.
- [ ] I used math evidence: numbers, an equation, or a model.
- [ ] I used at least two math words.

Writing Guide — Teacher Copy

- The answer to the math question is correct.
- The evidence is real lesson mathematics (numbers, equation, or model), not restated claim.
- The support level changed the language scaffolding, never the mathematical expectation.

Answer Key & Teacher Guide

1. **Notes 1:** Mixed Number
2. **Notes 2:** Improper Fraction
3. **Notes 3:** Convert
4. **Notes 4:** Simplify
5. **Notes 5:** Reciprocal
6. **Watch for this mistake:** *A common mistake in Divide Mixed Numbers is dividing the whole-number parts and fraction parts separately instead of converting first — for example, treating 4 1/2 ÷ 1 1/2 as (4 ÷ 1) + (1/2 ÷ 1/2) = 5. Always convert each mixed number to one improper fraction first, then divide using the reciprocal shortcut: 4 1/2 ÷ 1 1/2 = 9/2 ÷ 3/2 = 9/2 × 2/3 = 3.*
7. **Try It 1:** B. 2 — *3 ÷ 1 1/2 = 3 ÷ 3/2 = 3 × 2/3 = 6/3 = 2 sections.*
8. **Try It 2:** B. 3 — *7 1/2 = 15/2 and 2 1/2 = 5/2. So 15/2 ÷ 5/2 = 15/2 × 2/5 = 30/10 = 3 doorways.*